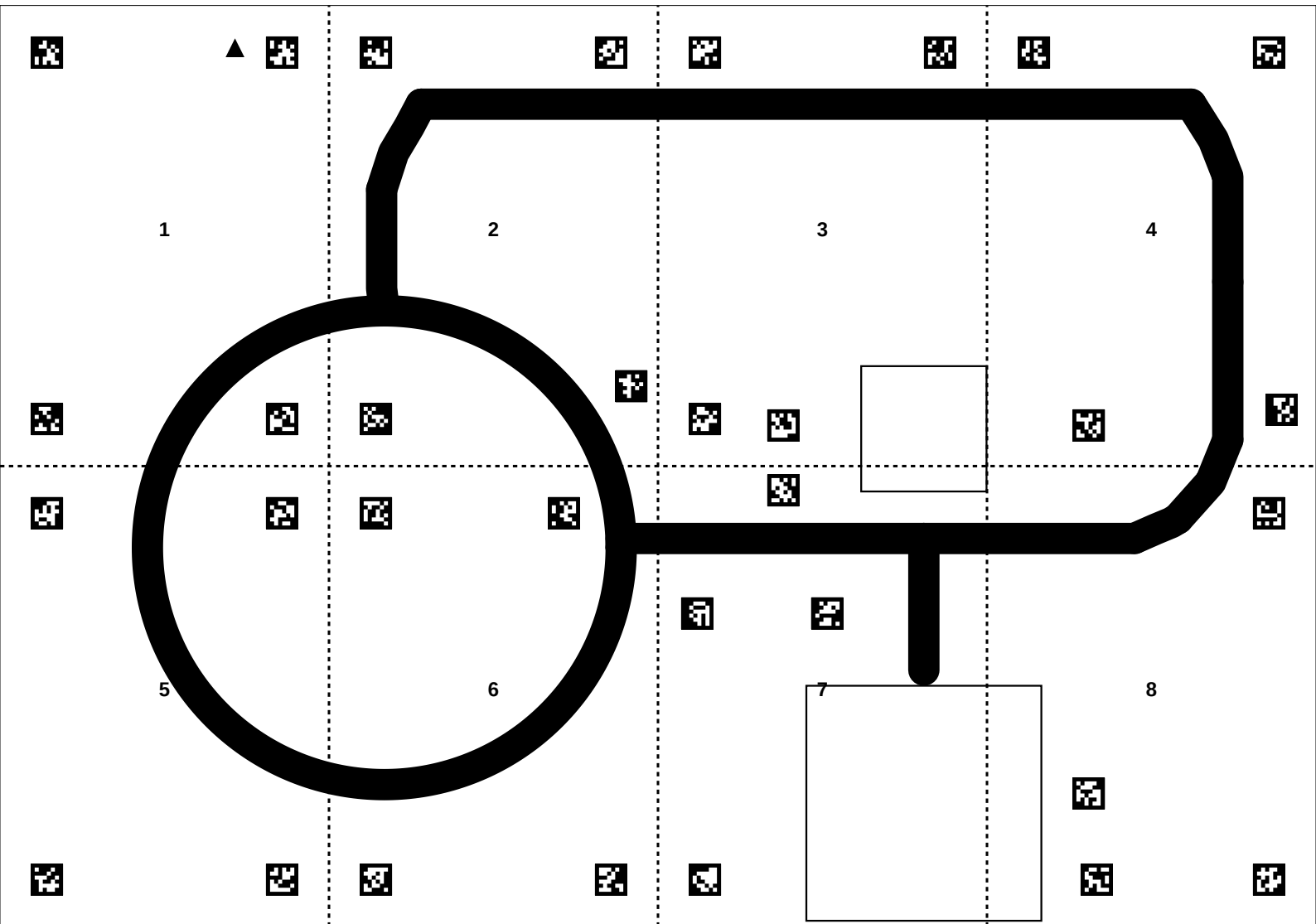


Task-1 Map — Overview

Full-map overview — tile layout reference



Assemble order

Top row (north): tiles 1 – 4, left to right.

Bottom row (south): tiles 5 – 8, left to right.

Align the corner crosses on adjacent tiles and tape on the back.

Task-1 Map — Print & Assemble Guide

1. Print the 8 map tiles (PDF pages 4–11)

Print PDF pages 4–11 on A4 at 100% / Actual Size. Do NOT use “fit to page”.

Verify the scale bar (X 80 mm / Y 40 mm) with a ruler before assembling.

2. Assemble (4 columns x 2 rows) — see the overview on PDF page 1

Top row (north) = tiles 1–4 left to right; bottom row (south) = tiles 5–8 left to right.

Align the corner crosses on adjacent tiles; tape on the back.

3. Orientation

The N arrow points map-north (up). Tags are printed with ID upright facing north.

4. AprilTag landmarks (the printed squares)

32 AprilTags (family 36h11, 20 mm) are printed onto the map as absolute-position landmarks.

Each is labelled “ID n” with an “N ↑” marker (yaw = 0 = facing map-north).

Web search: “AprilTag”, “AprilTag 36h11”, “AprilTag pose estimation”, “OpenCV aruco AprilTag”.

5. Scale

Map 840 x 588 mm = original 1000 x 700 mm Task-1 map scaled 0.84.

Route black line 20 mm; AprilTags 20 mm (physical sizes unchanged).

AprilTag Coordinates

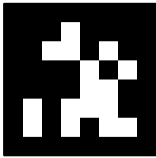
X and Y are in millimetres from the map's SOUTH-WEST corner (X east, Y north); NE corner = (840, 588). cm = mm ÷ 10.

ID	X (mm)	Y (mm)	ID	X (mm)	Y (mm)
0	30.0	558.0	16	30.0	264.0
1	180.0	558.0	17	180.0	264.0
2	30.0	324.0	18	30.0	30.0
3	180.0	324.0	19	180.0	30.0
4	240.0	558.0	20	240.0	264.0
5	390.0	558.0	21	360.0	264.0
6	240.0	324.0	22	240.0	30.0
7	403.0	345.0	23	390.0	30.0
8	450.0	558.0	24	445.0	200.0
9	600.0	558.0	25	500.0	278.5
10	450.0	324.0	26	450.0	30.0
11	500.0	320.0	27	528.0	200.0
12	660.0	558.0	28	695.0	85.0
13	810.0	558.0	29	810.0	264.0
14	695.0	320.0	30	700.0	30.0
15	818.0	330.0	31	810.0	30.0

Same values as scratch/landmarks/task1-tag-map.json (metres = mm ÷ 1000).



tile 1/8 ↑
ID 0



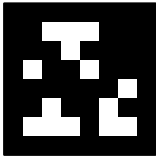
tile 1/8 ↑
ID 1



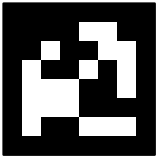
N



N ↑
ID 2



N ↑
ID 3



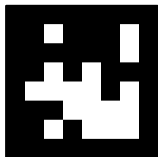
tile 1/8



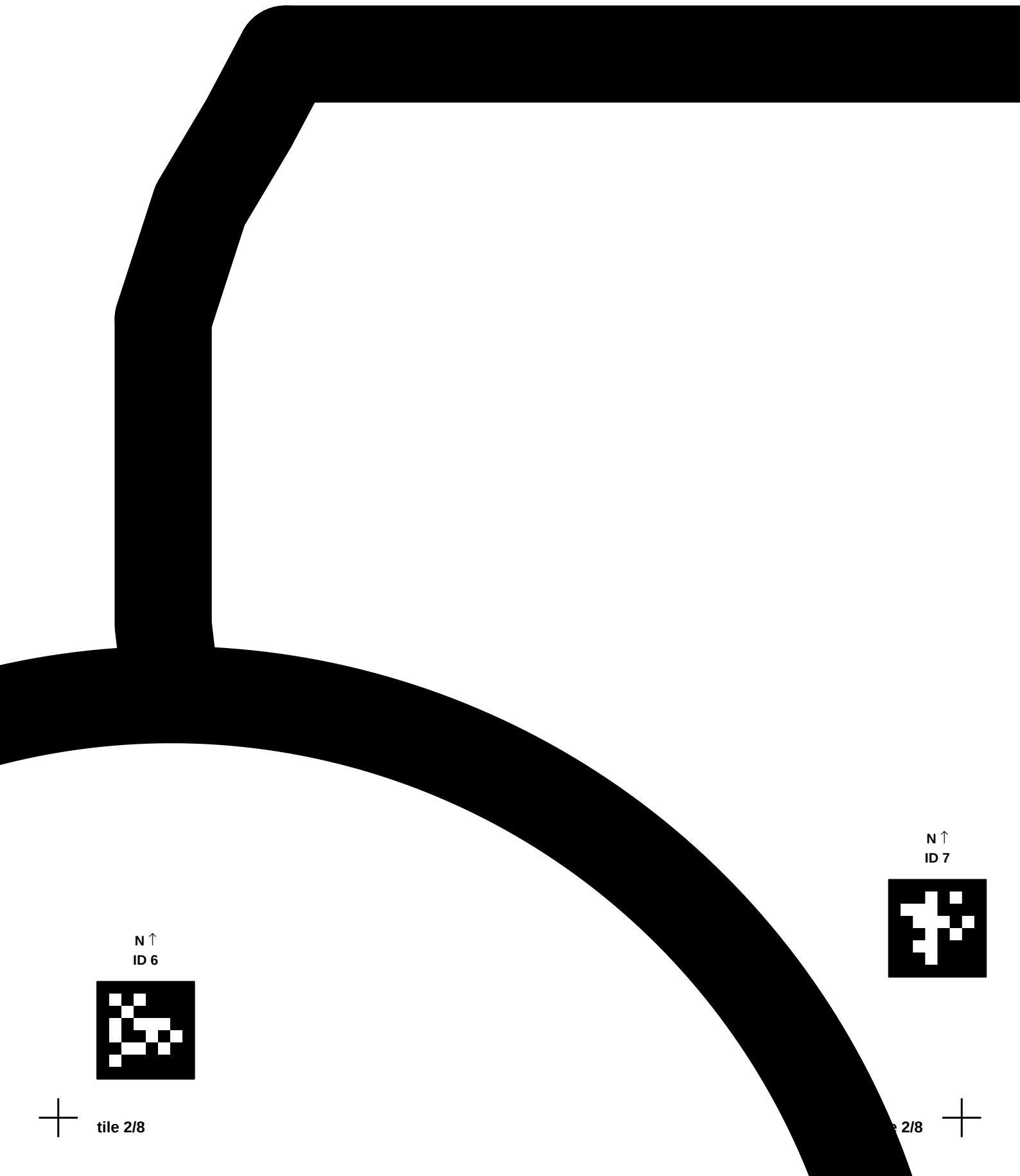
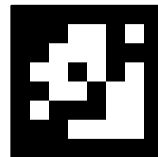
tile 1/8



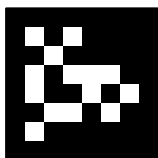
tile 2/8 ↑
ID 4



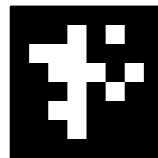
N tile 2/8
ID 5



N ↑
ID 6



N ↑
ID 7



tile 2/8



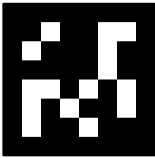
tile 2/8



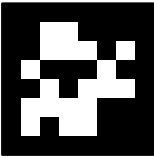
tile 3/8 ↑
ID 8



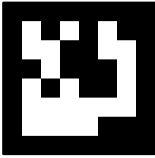
tile 3/8 ↑
ID 9



N ↑
ID 10



N ↑
ID 11



N ↑



tile 3/8

scale bar (X 80 / Y 40 mm)

40 mm

0

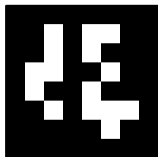
80 mm

tile 3/8

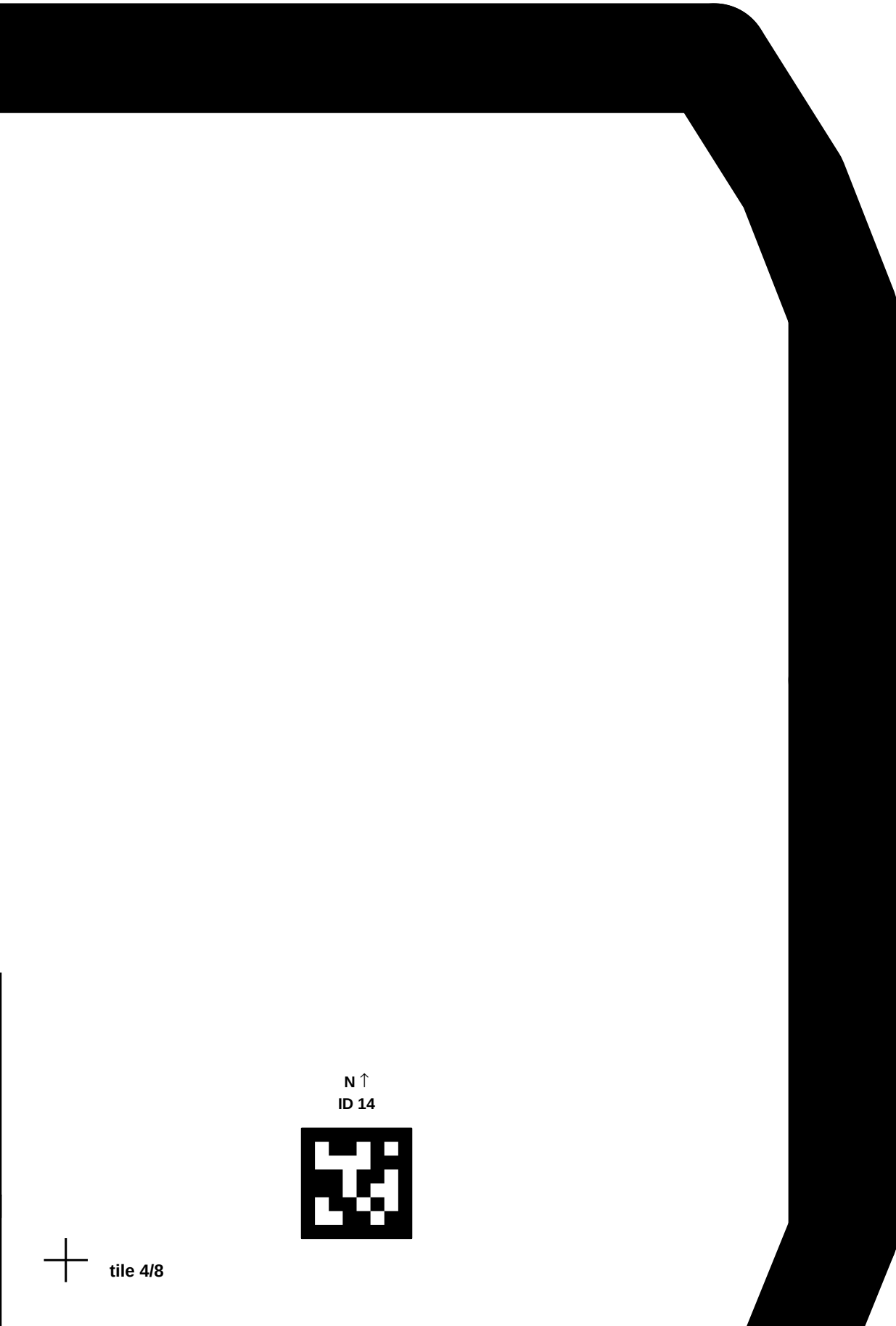
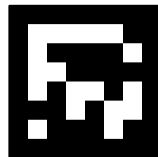




tile 4/8 ↑
ID 12



N tile 4/8
ID 13

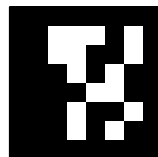


N ↑
ID 14



tile 4/8

N ↑
ID 15



tile 4/8

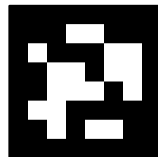




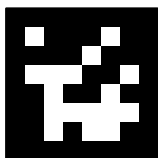
tile 5/8 ↑
ID 16



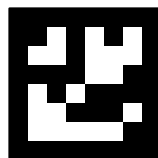
tile 5/8 ↑
ID 17



N ↑
ID 18



N ↑
ID 19



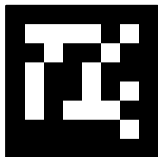
tile 5/8



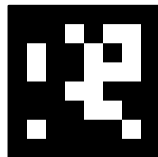
tile 5/8



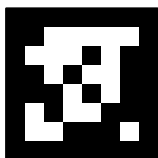
tile 6/8 ↑
ID 20



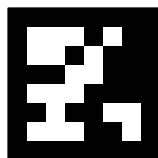
N ↑
ID 21



N ↑
ID 22



N ↑
ID 23



tile 6/8

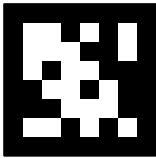


tile 6/8



tile 7/8

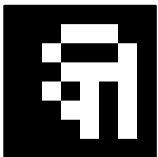
N ↑
ID 25



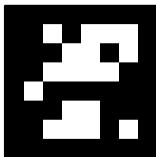
tile 7/8



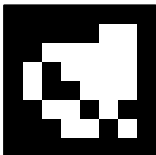
N ↑
ID 24



N ↑
ID 27



N ↑
ID 26



tile 7/8

tile 7/8



DEPARTURE AREA

START



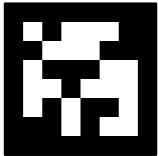
tile 8/8



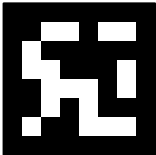
N tile 8/8
ID 29



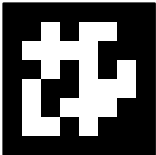
N ↑
ID 28



N ↑
ID 30



N ↑
ID 31



tile 8/8



tile 8/8